Chapter 4 Storage Exam questions

1. (Multi-Answer Question) Which of the following parameters are mandatory when FusionCompute connects to IP SAN storage?
   A. Storage resource type
   B. Storage management IP address
   C. Storage service port IP address
   D. Username and password

2. (Single Answer) It is a device, which manages the physical disk drives and presents them to the users as logical units. Which of the following device fits the above description?
   A. Host.
   B. Storage NIC.
   C. Storage Controller.
   D. Storage Cache.

3. (Single Answer) Which one of the following is a type of mechanical drive?
   A. SSD.
   B. SAS Drive.
   C. NVME Drive.
   D. Flash Drive.

4. (Multiple Answer) Which of the following are attached to the storage system **serially**?
   A. ATA Drive.
   B. SAS Drive.
   C. NL-SAS Drive.
   D. IDE Drive.

5. (Single Answer) Which of the following disks has the best performance in terms of speed?
   A. NL-SAS
   B. SAS
   C. SSD
   D. SATA

6. (T or F) SAN is a storage architecture that connects storage device and application servers through network.
   True

7. (Single Answer) Which of the following represents the logical storage resource connected to the target in SAN?
   A. LUN.
   B. Volume.
   C. Tape Drive.
   D. Disk Drive.

8. (Single Answer) Which of the following protocol has the slowest performance.
   A. SAS.
   B. FC.
   C. SATA.
   D. InfiniBand.

9. (Single Answer) Which of the following storage protocol allows **FC packets** to be transmitted **over Ethernet networks**?
   A. iSCSI.
   B. Ethernet.
   C. FC.
   D. FCoE.

10. (T or F) When configuring NAS file share on Linux systems using NFS, the shared directory do not need to be mounted.
   False

11. (Single Answer) Which of the following is **not a** common NAS protocol?
   A. FTP.
   B. CIFS.
   C. SSH.
   D. NFS.

12. SATA disks use parallel ports and SAS disks use serial ports.
   False

13. In RAID 5 and RAID 6, the number of faulty disks that can be tolerated is ( ), respectively.
   A. 1 , 2
   B. 2 , 3
   C. 2 , 1
   D. 3 , 2

14. The universal storage space unit mapped to the host in the disk array is ( ), which is the logical space created based on RAID.
   A. Storage pool
   B. Volume
   C. LUN
   D. Partition

15. Compared with SAN, which of the following is not an advantage of NAS?

   A. The file system of NAS supports a single file with higher specifications.
   B. NAS supports more files.
   C. Block data transmission is optimized.
   D. File sharing is optimized.

16. (multiple answers) Which of the following network file sharing protocols are supported by NAS?

A. FTP
B. CIFS
C. NFS
D. NTFS

17. （Multiple）Which of the following storage resource types are supported by Huawei FusionCompute?

A. FC SAN
B. IP SAN
C. NAS
D. Advanced SAN
E. FusionStorage

18. (Multiple）Which of the following are true about RAID?

A. A RAID group can be regarded as storage space consisting of two or more disks. The available capacity of a RAID group is the sum of the capacity of disks in the group.
B. RAID improves the I/O performance of a storage system by concurrently reading and writing data on multiple disks.
C. When a disk is damaged, other disks in RAID 5 can still work properly, and the disk I/O performance is not affected.
D. Compared with RAID 5, RAID 6 has stronger data recovery capability, but weaker read/write capability of disks.

19. （T or F）After the disks in an array form a RAID group, the server connected to the disk array cannot directly access the RAID group. Therefore, you need to divide the disk array into logical LUNs before allocating them to the server.

20. (Multiple）In Huawei FusionCompute, which of the following statements are true about adding shared storage for multiple hosts?

A. When the shared storage is IP SAN, make sure that the storage network of the host is connected to the service ports of storage.
B. Select the corresponding host when adding a datastore.
C. Select all hosts when associating storage devices.
D. Select all hosts when initializing storage resources

21. In Huawei FusionCompute, which of the following statements is true about adding storage ports to hosts?

A. storage ports can be added to hosts in batches.

B. select a host and add a storage port on the host configuration page.

C. The VLAN ID can be set to 0.

D. if FC SAN is used, users don’t need to add storage ports.

22. (Single）In Huawei FusionCompute, which of the following operations is supported when users modify a disk of a VM?

A. Change the disk mode from Dependent to Independent & persistent.

B. Change the disk capacity from 80 GB to 8 GB.

C. Change the disk status from Available to Unavailable.

D. Change the disk configuration mode from Thin provisioning to Ordinary.

23. Which functions are supported by thin provisioning?

A. virtual storage thin provisioning is irrelevant to the OS and hardware. Disk thin-provisioning can be provided as long as the virtual image management system is used.

B. this function enables disk space monitoring and reclaiming

C. this function enables alarms over data store usage. If the data usage exceeds the preset threshold, an alarm will be reported..

D. NTFS VMs do not support this function.

24. Which of the following about creating a new disk in **fusion is <u>false</u>**?

A. **Set disk mode to independent & non persistent and change the value of Raw Device Mapping to supported**

B. Set the disk configuration mode to thin provisioning and disk mode to independent & persistent

C. Set the disk configuration mode to thin provisioning and disk mode to independent & non persistent

D. Set the disk type to common and disk mode to independent & non persistent

25. VRM recommended to be deployed on which storage?

A. **Local disk**

B. FC SAN

C. IP SAN

D. NAS

26. Administrator can scan storage devices or manually refresh the the page to discover storage devices? **False**

27. Which storage types doesn't support disk capacity expansion?

A. **Non-virtualized local disk**

B. Virtual SAN

C. NAS

D. FusionStorage

28. Which statements about storage is false?

A. A physical storage device can serve as storage resource.
B. A storage device corresponds to a logical unit in physical storage .for example a LUN.
C. A data store can be IP SAN, FC SAN or NAS.
D. A VM disk file is in VHD format.

29. The resources in a cluster consist of hosts and associated storage and network resources. **True**

30. When use IP SAN storage, the steps to create disks include:

1. Create storage resource
2. Add storage ports on the host
3. Ad data stores
4. Create disks

The operation sequence is:
2-1-3-4

31. In the status of disk, if the used capacity is less than that of capacity. The possible reason is:

A. the disk type is shared
B. the disk mode is thin provisioning
C. the disk status is unavailable
D. the disk mode is dependent

32. After data in a VM is deleted, the disk size will not be automatically reduced. **True**

33. Which VM disk configuration mode need the least physical storage space?

A. Thin provisioning
B. Independent and persistent
C. Independent and non-persistent
D. Thick provisioning

34. Which are **true** about **shared** disks?

A. A disk of this type can be attached to multiple VM
B. No data store change for this type of disks
C. A disk of this type can be attached to only one VM
D. The VM with this type attached does not support snapshot function.

35. Given that:
    - When snapshot is created for the VM, no snapshot is created for the disk.
    - When a VM is restored using the snapshot, the disk is **not** restored and all data is written to the disk permanently.

    Which of the following disk modes should be selected?
    A. Dependent
    B. Independent
    C. Independent and persistent
    D. Independent and non-persistent

36. For VMs that used RDM storage which **are not supported**?
    A. Snapshot
    B. Thin provisioning
    C. Storage live migration
    D. CPU hot-add

37. Which is true about CIFS?
    A. CIFS supports multi-user access
    B. CIFS needs dedicated software install
    C. CIFS has low requirements on network transmission reliability
    D. In CIFS, fault recovery is available

38. Which is false about iSCSI an fiber channel?
    The answer: **the transmission distance of FC is longer than the TCP/IP**

39. In Huawei FusionCompute, ( ) can only be used as **non-virtualized** storage.
    A. SAN
    B. NAS
    C. Local disk
    D. Fusion Storage

40. Which is **true** about virtualized storage and non-virtualized storage?
    A. Virtualized storage is more secure than non-virtualized storage
    B. The read and write performance of non-virtualized storage is better than that of virtualized storage.
    C. The scalability of non-virtualized storage is better than that of virtualized storage.
    D. The compatibility of virtualized storage is better than that of non-virtualized storage.

41. In Linux VM, ( ) is used to manage logical volumes.
    A. DVM
    B. LVM
    C. PVM
    D. OVM

42. In Huawei FusionCompute, you can perform which of the following operations on the VRM?
    A. Divide RAID groups for disk arrays.
    B. Manually add a storage device to a VM.
    C. Add SAN storage as non-virtualized storage
    D. Convert virtualized storage into non-virtualized storage.

43. The correct procedure for deleting storage resources is:
    1. Delete or migrate the VM disk
    2. Disassociate the host.
    3. Delete the storage resources.
    4. Destruct the datastore.

The answer is : 1→4→2→3

44. In Huawei FusionCompute, which of the following storage virtualization features are supported?
    A. Thin provisioning
    B. Storage live migration
    C. Snapshot
    D. Disk capacity expansion

45. Storage virtualization can be classified into.
    A. Block-level virtualization
    B. Page-level virtualization
    C. File-level virtualization
    D. Application-level virtualization

46. SATA disks use parallel ports and SAS disks use serial ports. **False**

47. SSD are **mechanical** hard disks, featuring fast read/write speed, low power consumption, and lightweight. However, SSDs have limited service life. **False**

48. Based on RAID 5, RAID 6 enhances the fault tolerance capability of disks and allows two disks to be faulty at the same time. The minimum number of disks in a RAID 6 group is 4. **True**

49. RAID 1 can replace data backup. Data mirroring is used to improve the fault tolerance of disk arrays. The data recovery cost is the smallest among all RAID levels. **False**

50. Which RAID level has the highest disk space usage?
    A. RAID 0
    B. RAID 1
    C. RAID 5
    D. RAID 6

51. Which RAID level has the weakest data redundancy capability?
    A. **RAID 0**
    B. RAID 1
    C. RAID 5
    D. RAID 6

52. Which of the following is recommended for the **archive room** where a large amount of data needs to be stored for a long time and doesn't need to be read frequently?
    A. SSD disk
    B. SAS disk
    C. SATA disk
    D. NL-SAS disk

53. Common categories of disk arrays include.
    A. RAID
    B. JBOD(just bunches of disks)
    C. SBOD(Switched Bunches of Disks)
    D. MAID(Massive Array of Idle Disks)

54. The common **external** storage solutions include().
    A. DAS
    B. SAN
    C. NAS
    D. NFS

55. Compared with FC SAN, which of the following is not an advantage of IP SAN?
    A. Low construction cost
    B. Long transmission distance
    C. High transmission efficiency
    D. Convenient bandwidth expansion

56. Which is not included in NAS architecture?
    A. Status verification module
    B. File system
    C. Volume management module
    D. Network port

57. iSCSI is a standard protocol based on the ( ) protocol and SCSI instructions set. It is widely used in many storage networks of the ( ) architecture.
    A. FC, FC SAN
    B. TCP/IP , FC SAN
    C. TCP/IP, IP SAN
    D. FC, IP SAN

58. Which of the following is false about difference between SAN and NAS?
    A. NAS has its own file system, but SAN does not.
    B. SAN provides block-level data transmission and storage services, while NAS provides file-level transmission and storage services.
    C. SAN is a network architecture, while NAS is a network file storage device.
    D. SAN can expand storage space, but NAS devices cannot be expanded.

59. In a NAS environment, data inconsistency may occur when multiple hosts write data to one file at the same time. **True**

60. NAS has its own file system. After accessing NAS through the network sharing protocol, the application server can read and write files in the file system. **False**

61. SAN provides a host with raw disk where no file system is created. The file system to be created depends on the operating system of the host. **True**

62. When configuring IP SAN for a host. You need to configure ( ) for the host on the storage device to map the host to the IP SAN.
    A. iSCSI Server
    B. iSCSI Client
    C. iSCSI Initiator
    D. iSCSI Target

63. Which is **true** about **CIFS**?
    A. CIFS has low requirements on network transmission reliability. The TCP/IP or UDP protocol can be used.
    B. CIFS requires the client to have a dedicated software installed
    C. CIFS supports multi-user access.
    D. CIFS retains the file format characteristic

64. Which disk format is not supported by Huawei FusionCompute?
    A. VMDK
    B. QCOW2
    C. RAW
    D. VHD

65. In Huawei FusionCompute, which is **false** about raw device mapping (RDM)?
    A. RDM disks deliver the best performance among all disk types
    B. An RDM datastore corresponds to only one disk.
    C. It is recommended that the OS to be installed on an RDM disk
    D. An RDM datastore can only correspond to a LUN on SAN storage

66. Which of the following is true about distributed storage?
    A. Disks of distributed storage are from x86 disk arrays.
    B. Disks of distributed storage are from x86 VMs.
    C. Disks of distributed storage are from x86 servers.
    D. Disks of distributed storage are from x86 laptops.

67. Fusion Storage can be used as a virtualized storage device. **False**

68. Which of the following is false about the **figure 4-26 OS file system in page 94**.
    A. Files are stored in the file system and mapped to the file system block by the file system.
    B. The file system block corresponds to the logical extent of LVM.
    C. The logical extent is mapped to the physical extent of the disk through LVM.
    D. Each file system block corresponds to discontinues extents on the physical disk.

69. Which is true about NFS?
    A. No special software needs to be installed on the NFS client.
    B. NFS is a state full protocol
    C. NFS support multi- user access
    D. The user permission cannot be set for NFS.

70. The process of creating a file system on a disk is usually called ().
    A. RAID
    B. Partition
    C. Formatting
    D. Encapsulation

71. Disk partitions are divided by the OS on physical disks. **True**

72. In FusionCompute, system disks of VMs cannot be detached. **True**

73. Which is **false** about the VIMS file system?
    A. VIMS is a high-performance cluster file system
    B. VIMS is the technical basis of advanced features such as thin provisioning disk, snapshot, and storage migration.
    C. VIMS is compatible with FC SAN, IP SAN and local disks.
    D. VIMS is a universal storage virtualization file system in the industry.

74. The VM disk file format on FusionCompute is
    A. OVF
    B. VMDK
    C. VHD
    D. DHT

RAID Info:

1. **RAID 0 highest storage**
2. **RAID 1**
    i. **Best data security and availability**
    ii. **Lowest disk usage**
    iii. **Most expensive**
3. **RAID 5 is the optimal**
4. **RAID 6 used in environment requires high security, accuracy and integrity.**